

Common Mistakes and How to Fix Them: Optimization

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Introduction

These notes summarize common mistakes students make when learning optimization techniques such as Gradient Descent, Ridge Regression, SGD, and Momentum. For each mistake, you will find an explanation of why it happens and a clear discussion of how to fix it, with concrete mathematical reasoning and small examples.

Common Mistakes

1. Thinking the Learning Rate Depends Mainly on Data Size

What the mistake looks like: Choosing a step size because “the dataset is large enough” without checking $\lambda_{\max}(X^T X)$. **Why it’s easy to make:** Dataset size often matters in algorithms, so it’s natural to think it controls stability.

How to fix it: GD stability comes from directional behavior. In direction $v_{\max}$ (the eigenvector of the largest eigenvalue), the update multiplies the component by $(1 - 2\eta\lambda_{\max})$. If $\eta \geq 1/\lambda_{\max}$, this term becomes negative or too large in magnitude, causing divergence or oscillation. Dataset size never appears in this expression. **Example:** Incorrect: $\eta = 1$ “because n is large.” Correct: If $\lambda_{\max} = 100$, then we need $\eta < 0.01$ to make $(1 - 2\eta\lambda_{\max})$ lie in $(-1, 1)$.

2. Assuming Ridge Shrinks All Directions Equally

What the mistake looks like: Thinking Ridge multiplies the solution by some constant like $1/(1 + \lambda)$. **Why it’s easy to make:** The term “weight decay” sounds uniform.

How to fix it: In the SVD basis, the coefficient in direction i becomes

$$\frac{\sigma_i}{\sigma_i^2 + \lambda}(u_i^T y).$$

When σ_i is tiny, the factor is almost 0. When σ_i is large, the factor is roughly $1/\sigma_i$. Since each singular direction has a different σ_i , shrinkage is unequal. **Example:** Incorrect: Applying one global scaling. Correct: If $\sigma_1 = 10$ and $\sigma_2 = 0.01$ with $\lambda = 1$, then $\frac{10}{100+1} \approx 0.099$ but $\frac{0.01}{0.0001+1} \approx 0.009999$, nearly zero.

3. Expecting GD to Move Equally Fast in All Directions

What the mistake looks like: Believing low-variance directions converge quickly. **Why it’s easy to make:** GD feels like a single “speed” update.

How to fix it: In direction i ,

$$\tilde{w}_{t+1,i} = (1 - 2\eta\sigma_i^2)\tilde{w}_{t,i} + 2\eta\sigma_i(u_i^T y).$$

If σ_i is small, then $2\eta\sigma_i^2$ is extremely tiny. That means GD barely updates that component. **Example:** Incorrect: “GD updates everything at the same rate.” Correct: If $\sigma_i = 0.01$ and $\eta = 0.1$, then $1 - 2\eta\sigma_i^2 \approx 0.99998$, almost unchanged per step.

4. Thinking Early Stopping Has No Connection to Ridge

What the mistake looks like: Treating early stopping as merely saving compute. **Why it's easy to make:** Ridge changes the objective; stopping time feels unrelated.

How to fix it: High- σ_i directions converge quickly; low- σ_i directions take far longer. Stopping early means the low- σ_i components never grow, effectively suppressing them just as Ridge does. **Example:** Incorrect: Stopping at $t = 50$ “just stops training.” Correct: At $t = 50$ you may have nearly converged in high-variance directions but barely moved in low-variance ones—selective shrinkage identical in effect to Ridge.

5. Believing Noisy Mini-Batch Gradients Are Biased

What the mistake looks like: Assuming randomness means the gradient estimate is “wrong.”

Why it's easy to make: Noise is often associated with error.

How to fix it: Uniform sampling gives:

$$\mathbb{E}[\nabla f_i(\theta)] = \frac{1}{n} \sum_{i=1}^n \nabla f_i(\theta) = \nabla f(\theta).$$

Noise only increases variance, not bias. Over many steps, the expectation matches the true gradient. **Example:** Incorrect: “SGD optimizes a different function.” Correct: The average gradient over random samples equals the full gradient.

6. Thinking Momentum Just Increases Step Size

What the mistake looks like: Treating βz_k as amplification. **Why it's easy to make:** Momentum uses a new variable, so it looks like a scaled step.

How to fix it: Expanding the recurrence,

$$z_k = (1 - \beta) \sum_{i=0}^{k-1} \beta^{k-1-i} \nabla f(w_i),$$

shows an exponential average of past gradients. Opposing gradients partially cancel, reducing oscillation; consistent gradients accumulate and speed convergence. **Example:** Incorrect: “Momentum just overshoots.” Correct: If gradients alternate sign, the weighted average damps them, preventing bouncing.

7. Believing SGD Needs a Decaying Learning Rate to Converge

What the mistake looks like: Insisting constant-step SGD “must diverge.” **Why it's easy to make:** Classical theory often requires step-size decay.

How to fix it: The convergence proof shows a contraction on expected loss:

$$\mathbb{E}[L(z_{t+1})] \leq (1 - C)L(z_t).$$

As long as such a contraction holds, a constant η works. The key is that the proof tracks loss, not the parameters themselves. **Example:** Incorrect: “Loss can't decrease with constant η .” Correct: If every step reduces expected loss by a fixed proportion, geometric convergence follows.